

EXAMPLE_PROJECTS_INDEX

Example Projects Analysis - Complete Index

Overview

This analysis covers three production-grade AI-powered CLI projects, providing comprehensive documentation for porting features to HelixCode. The analysis spans 4,500+ words across three detailed documents.

Analysis Date: November 6, 2025
Projects Analyzed: Qwen Code, Gemini CLI, DeepSeek CLI
Total Files Examined: 50+ files across source code and documentation

Generated Documents

1. EXAMPLE_PROJECTS_ANALYSIS.md (20KB, 2,696 words)

Comprehensive technical analysis of all three projects

Contains: - Detailed overview of each project - Core features and capabilities breakdown - Supported LLM providers and authentication methods - Complete list of supported models - Key technical implementations with code examples - Unique features suitable for porting - Session commands and CLI features - Comparative analysis table - Feature porting priorities - Model-specific integration patterns - Implementation roadmap for HelixCode - Key learnings and best practices

Best for: Deep technical understanding and implementation details

2. EXAMPLE_PROJECTS_QUICK_REFERENCE.md (7.8KB, 1,024 words)

Quick lookup guide with essential information

Contains: - Project overview comparison table - Core features summary for each project - Porting priority matrix (High/Medium/Low) - Implementation patterns for authentication and configuration - Key technical details for each project - File

location reference - Recommended import order - Configuration structure
examples - Testing strategy for each project - Deployment considerations

Best for: Quick lookups and decision-making during implementation

3. ANALYSIS_SOURCES.md (7KB, 757 words)

Complete documentation of sources and analysis methodology

Contains: - All files examined for Qwen Code - All files examined for Gemini CLI -
All files examined for DeepSeek CLI - Analysis methodology and approach -
Feature extraction summary - Key code locations for reference -
Recommendations for HelixCode integration phases - Document status and
metadata

Best for: Understanding what was analyzed and where to find source code

Quick Navigation

By Use Case

Getting Started with HelixCode Integration? 1. Start with
EXAMPLE_PROJECTS_QUICK_REFERENCE.md 2. Review “Recommended Import
Order” section 3. Follow Phase 1-5 roadmap in
EXAMPLE_PROJECTS_ANALYSIS.md

Need Implementation Details? 1. Consult EXAMPLE_PROJECTS_ANALYSIS.md
section 4 (Technical Implementations) 2. Use ANALYSIS_SOURCES.md for exact file
locations 3. Reference the source files directly

Looking for Specific Feature? 1. Use
EXAMPLE_PROJECTS_QUICK_REFERENCE.md feature matrix 2. Look up in
EXAMPLE_PROJECTS_ANALYSIS.md unique features sections 3. Find
implementation details in relevant project section

By Project

For Qwen Code Information: - Main section: EXAMPLE_PROJECTS_ANALYSIS.md
§ 1 - Quick ref: EXAMPLE_PROJECTS_QUICK_REFERENCE.md (Qwen Code section)
- Sources: ANALYSIS_SOURCES.md § Qwen Code Sources

For Gemini CLI Information: - Main section: EXAMPLE_PROJECTS_ANALYSIS.md
§ 2 - Quick ref: EXAMPLE_PROJECTS_QUICK_REFERENCE.md (Gemini CLI section)
- Sources: ANALYSIS_SOURCES.md § Gemini CLI Sources

For DeepSeek CLI Information: - Main section:
EXAMPLE_PROJECTS_ANALYSIS.md § 3 - Quick ref:
EXAMPLE_PROJECTS_QUICK_REFERENCE.md (DeepSeek CLI section) - Sources:
ANALYSIS_SOURCES.md § DeepSeek CLI Sources

Key Findings Summary

Three Distinct Approaches

1. **Qwen Code** - Enterprise with Multimodal Focus
 - Vision model auto-switching
 - DashScope token caching
 - Session token limits
 - Qwen OAuth authentication
2. **Gemini CLI** - Enterprise with Advanced Features
 - Google Search grounding
 - Full checkpointing system
 - Thinking mode with limits
 - Loop detection and compression
 - MCP protocol support
3. **DeepSeek CLI** - Local-First with Simplicity
 - Dual local/cloud architecture
 - Ollama integration
 - Minimal dependencies (5 core)
 - Setup automation
 - Simple, clean codebase

High-Priority Features for HelixCode

1. Vision model auto-switching (Qwen)
2. Google Search grounding (Gemini)
3. MCP Protocol support (Qwen/Gemini)
4. Token caching strategies (Qwen/Gemini)
5. Checkpointing system (Gemini)

Implementation Sequence

Phase 1 (Foundation): DeepSeek → Qwen → Gemini providers

Phase 2 (Advanced): Vision switching, caching, fallback logic

Phase 3 (Integration): MCP, Search, Checkpointing

Phase 4 (Enterprise): Loop detection, compression, IDE context

Phase 5 (Deployment): Automation and telemetry

Document Interconnections

EXAMPLE_PROJECTS_QUICK_REFERENCE.md

- └ Links to detailed sections in ANALYSIS.md
- └ References file locations in ANALYSIS_SOURCES.md
- └ Provides quick decision matrix

EXAMPLE_PROJECTS_ANALYSIS.md

- └ Detailed version of QR concepts
- └ Code examples from files listed in SOURCES.md
- └ Implementation roadmap guides QUICK_REFERENCE priorities

ANALYSIS_SOURCES.md

- └ Exact paths for all examined files
 - └ Supports deep dives in ANALYSIS.md
 - └ Cross-references to QUICK_REFERENCE
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Usage Recommendations

For Architecture Decisions

1. Review EXAMPLE_PROJECTS_QUICK_REFERENCE.md comparison tables
2. Check EXAMPLE_PROJECTS_ANALYSIS.md implementation patterns
3. Consult ANALYSIS_SOURCES.md for specific file examples

For Feature Implementation

1. Identify feature in QUICK_REFERENCE priority matrix
2. Read detailed implementation in ANALYSIS.md
3. Locate source code using ANALYSIS_SOURCES.md
4. Review actual code in cli_agents/

For Integration Planning

1. Follow roadmap in ANALYSIS.md (5 phases)
2. Use import order in QUICK_REFERENCE.md
3. Reference test strategies in QUICK_REFERENCE.md
4. Check deployment considerations

For Documentation Writing

1. Review project architectures in ANALYSIS.md
2. Check authentication patterns in QUICK_REFERENCE.md

3. Use key learnings from ANALYSIS.md
 4. Reference file locations in ANALYSIS_SOURCES.md
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Content Breakdown

EXAMPLE_PROJECTS_ANALYSIS.md Sections

1. Qwen Code overview and details (400 words)
2. Gemini CLI overview and details (450 words)
3. DeepSeek CLI overview and details (350 words)
4. Comparative analysis (300 words)
5. Implementation roadmap and learnings (200 words)

EXAMPLE_PROJECTS_QUICK_REFERENCE.md Sections

1. Project overview table (100 words)
2. Core features by project (250 words)
3. Porting priority matrix (200 words)
4. Implementation patterns (150 words)
5. Technical details and locations (300 words)

ANALYSIS_SOURCES.md Sections

1. Qwen Code sources (150 words)
 2. Gemini CLI sources (200 words)
 3. DeepSeek CLI sources (150 words)
 4. Methodology and extraction (150 words)
 5. Recommendations and index (107 words)
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Quick Links to Source Files

Qwen Code Key Files

- Models: `cli_agents/qwen-code/packages/cli/src/ui/models/availableModels.ts`
- DashScope: `cli_agents/qwen-code/packages/core/src/core/openaiContentGenerator/provider/dashscope.ts`
- Config: `cli_agents/qwen-code/README.md` (session config section)

Gemini CLI Key Files

- Models: `cli_agents/gemini-cli/packages/core/src/config/models.ts`
- Client: `cli_agents/gemini-cli/packages/core/src/core/client.ts`
- Auth: `cli_agents/gemini-cli/packages/core/src/code_assist/`

DeepSeek CLI Key Files

- API: `cli_agents/deepseek-cli/src/api.ts`
- CLI: `cli_agents/deepseek-cli/src/cli.ts`
- Interactive: `cli_agents/deepseek-cli/src/commands/interactive.ts`

Document Statistics

Document	Size	Words	Sections	Code Examples
ANALYSIS.md	20KB	2,696	20+	15+
QUICK_REFERENCE.md	7.8KB	1,024	15+	8+
ANALYSIS_SOURCES.md	7KB	757	10+	2+
Total	34.8KB	4,477	45+	25+

How to Use These Documents

Scenario 1: I want to add Qwen support to HelixCode

1. Read: `EXAMPLE_PROJECTS_QUICK_REFERENCE.md` (Qwen section)
2. Study: `EXAMPLE_PROJECTS_ANALYSIS.md` § 1 (Qwen Code)
3. Implement: Review files in `ANALYSIS_SOURCES.md` § Qwen Code Sources
4. Test: Follow testing strategy in `QUICK_REFERENCE.md`

Scenario 2: I need to implement vision model switching

1. Read: `EXAMPLE_PROJECTS_ANALYSIS.md` § 1.5 (Unique Features)
2. Review: `QUICK_REFERENCE.md` (Vision Model Auto-Switching)
3. Study: `Qwen_Code/packages/cli/src/ui/models/availableModels.ts`
4. Implement: Following the three-mode pattern

Scenario 3: I’m designing the LLM provider architecture

1. Compare: `QUICK_REFERENCE.md` (Project Overview table)

2. Study: ANALYSIS.md § 1-3 (Technical Implementations)
3. Reference: ANALYSIS.md (Model-Specific Integration Patterns)
4. Plan: Implementation Roadmap in ANALYSIS.md

Scenario 4: I want to understand authentication patterns

1. Review: QUICK_REFERENCE.md (Authentication Pattern section)
 2. Study: ANALYSIS.md § 1.2, 2.2, 3.2 (Provider sections)
 3. Find: Specific files in ANALYSIS_SOURCES.md
 4. Implement: Following the fallback pattern
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Next Steps

After reviewing these documents:

1. **Familiarize** with each project by reading the appropriate analysis section
 2. **Prioritize** features using the porting priority matrix
 3. **Design** your architecture based on implementation patterns
 4. **Review** source code using file location references
 5. **Implement** following the 5-phase roadmap
 6. **Test** using the testing strategies outlined
 7. **Deploy** considering deployment considerations
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Additional Resources

- Source projects located in: `/Users/milosvasic/Projects/helix_code/cli_agents/`
- Qwen Code: `cli_agents/qwen-code/docs/`
- Gemini CLI: `cli_agents/gemini-cli/docs/`
- DeepSeek CLI: `cli_agents/deepseek-cli/README.md`

All three projects have comprehensive documentation in their respective directories.

Document Maintenance

These documents are snapshots of analysis performed on November 6, 2025. As the example projects evolve, consider:

1. Re-running analysis on major version updates
2. Updating implementation patterns as best practices evolve
3. Adding new features to the porting priority matrix

4. Extending code examples as projects add capabilities

Last Updated: November 6, 2025